

Problem Statement Worksheet (Hypothesis Formation)

How can healthcare data be used to predict whether a patient is likely to have diabetes based on diagnostic health measurements, in order to support earlier identification and intervention?

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1 Context

Diabetes is a common chronic disease that can lead to serious health complications if not identified early. This project analyzes patient health measurements such as glucose level, BMI, blood pressure, insulin, age, and pregnancy history to identify patterns associated with diabetes and support early prediction of the condition.

2 Criteria for success

Success for this project means developing a machine learning model that can reasonably predict whether a patient has diabetes based on available health measurements.

The project should also identify which health factors appear most strongly associated with diabetes outcomes.

3 Scope of solution space

- Analyze relationships between patient health indicators and diabetes outcome.
- Clean and preprocess the dataset by handling invalid and missing values.
- Explore feature patterns through visualization and exploratory data analysis.
- Build and evaluate classification models for diabetes prediction.

4 Constraints within solution space

- The dataset contains missing or invalid values represented as zeros.
- Some features, such as insulin and skin thickness, contain a large number of missing observations.
- The dataset is relatively small (768 records), which may limit model generalization.
- The dataset only represents a specific patient population and may not generalize to all populations.

5 Stakeholders to provide key insight

- Healthcare providers and clinicians
- Public health researchers
- Patients at risk for diabetes
- Healthcare analytics and data science teams

6 Key data sources

- Pima Indians Diabetes Dataset
- CSV dataset containing patient diagnostic measurements and diabetes outcomes

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